

CRANBROOK SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2001

MATHEMATICS

3 UNIT (Additional)
4 UNIT (First Paper)

Time allowed – Two hours

DIRECTIONS TO CANDIDATES

- * Attempt all questions.
- * ALL questions are of equal value.
- * All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- * Standard integrals are printed on the back page.
- * Board-approved calculators may be used.
- * You may ask for extra Writing Booklets if you need them.

* Submit your work in four booklets :

- (i) QUESTION 1 (4 page)
- (ii) QUESTIONS 2 & 3 (8 page)
- (iii) QUESTIONS 4 & 5 (8 page)
- (iv) QUESTIONS 6 & 7 (8 page)

1. (4 page booklet)

- (a) Evaluate $\int_0^{\pi/2} \cos^2 x \, dx$ [2 marks]
- (b) (i) On the same set of axes, sketch the graphs of $y = 2|x|$ and $y = |x-3|$
(ii) Hence or otherwise solve for x $2|x| \leq |x+3|$ [4 marks]
- (c) In an Arithmetic Sequence, whose first term and common difference are both non-zero, T_n represents the n^{th} term and S_n represents the sum of the first n terms. Given that T_6, T_9, T_{10} form a Geometric Sequence
(i) show that $S_{10} = 0$
(ii) show that $S_6 + S_{12} = 0$
(iii) deduce that $T_7 + T_8 + T_9 + T_{10} = T_{11} + T_{12}$ [6 marks]

2. (new 8 page booklet please)

- (a) Evaluate
(i) $\sin^{-1}\left(\frac{1}{2}\right)$ (ii) $\sin^{-1}\left(\cos \frac{\pi}{3}\right)$ [2 marks]
- (b) State the Domain and Range of $y = \sin^{-1}(1-x^2)$ [2 marks]
- (c) Sketch the graphs of (i) $y = \sin^{-1}x + \cos^{-1}x$
(ii) $y = \sin^{-1}(1-x)$ [4 marks]
- (d) Find the exact volume of the solid of rotation when the area bounded by the curve $y = \frac{1}{\sqrt{1+4x^2}}$ and the x -axis from $x = -\frac{1}{2}$ to $x = \frac{1}{2}$ is rotated about the x -axis. [4 marks]

3.

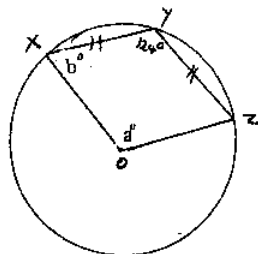
- (a) (i) Show that $(x-2)$ is a factor of $4x^3 - 8x^2 - 3x + 6$.
(ii) Find the general solution of $4\sin^2 \theta - 8\sin^2 \theta - 3\sin \theta + 6 = 0$. [4 marks]
- (b) Given $\sin \theta = \frac{4}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$ find $\sin 2\theta$. [2 marks]
- (c) Show that $\frac{\sin 3\phi}{\sin \phi} - \frac{\cos 3\phi}{\cos \phi} = 2$. [3 marks]
- (d) Using the transformation $R \sin(x+\alpha)$ solve $\sqrt{3} \sin x + \cos x = 1$ for $-\pi \leq x \leq \pi$. [4 marks]

4. (new 8 page booklet please)

- (a) Find the locus of $M(x, y)$ in cartesian form given :
 $x = p + q$
 $y = \frac{1}{2}(p^2 + q^2 + 4)$
 and $pq = 2$ [2 marks]
- (b) A is the fixed point $(-4, 8)$. P is a variable point on the parabola $x^2 = 8y$. Prove that the locus of M, the midpoint of AP, is a parabola with vertex $(-2, 4)$ and focal length 1 unit. [5 marks]
- (c) (i) Explain why $e^x - 2x - 1 = 0$ must have a root between 1.2 and 1.3
 (ii) By using Newton's method (twice), and taking 1.3 as a first approximation, find a better approximation to the root, giving your answer correct to three decimal places. [5 marks]

5.

- (a) In the diagram shown, $XY = YZ$ and O is the centre of the circle.
 $\angle XYZ = 124^\circ$
 Evaluate a and b, giving reasons for your answers.



[3 marks]

- (b) Points A, B, C and D lie on a circle such that chords BC and CD are equal and AD is a diameter of the circle (B and C are in the same half of the circle). BX is drawn parallel to CD, meeting AD in X.
 (i) Draw a neat and clear diagram representing the situation.
 (ii) Let $\angle CDB = x^\circ$. Prove that ABX is an isosceles triangle. [5 marks]
- (c) Two of the roots of the equation $x^3 + ax^2 + b = 0$ are reciprocals of each other.
 (i) Show that the third root is equal to $-b$.
 (ii) Show that $a = b - \frac{1}{b}$ [4 marks]

3

6. (new 8 page booklet please)

- (a) The daily growth rate of a population of a species of mosquito is proportional to the excess of the population over 5000
 i.e. $\frac{dP}{dt} = k(P - 5000)$.
 (i) Show that $P = 5000 + Ae^{kt}$ is a solution of this differential equation. [2 marks]
 (ii) If initially $P = 5002$ and after 6 days the population is 25000 find the values of A and k in exact form. [2 marks]
 (iii) Find the mosquito population after 10 days (to the nearest whole number). [2 marks]
- (b) On a certain day in July, 2001 the depth of water at high tide over a harbour bar in Auckland was $10\frac{2}{3}$ m and at low tide $6\frac{1}{4}$ hours earlier it was 7m. High tide occurred at 3.40 p.m. on this day.
 (i) Assuming that the tide's motion is simple harmonic and of the form $x = -n^2(x - b)$, where $x = b$ is the centre of motion and $x = a$ is the amplitude, show that $x = b - a \cos nt$ satisfies this equation for simple harmonic motion. [2 marks]
 (ii) Hence or otherwise find the earliest time before 3.40 p.m. on this day at which a ship requiring a $9\frac{1}{2}$ m depth of water could have crossed the bar (to the nearest minute). [4 marks]

7.

- (a) Prove by mathematical induction that $3^n + 7^n$ is always even for n a positive integer. [5 marks]
- (b) An executive borrows \$P at r% per fortnight reducible interest and pays it off at \$F per fortnight in n equal fortnightly instalments. (Assume that there are 26 fortnights in one year.)
 (i) If D_n is the debt remaining after n fortnights prove that

$$D_n = P \left(1 + \frac{r}{100}\right)^n - F \times \left[\frac{\left(1 + \frac{r}{100}\right)^n - 1}{\frac{r}{100}} \right]$$
 [3 marks]
 (ii) If $D_n = 0$ prove that $n = \frac{\log_e \left[\frac{F}{F - rP} \right]}{\log_e \left(1 + \frac{r}{100} \right)}$ [2 marks]
 (iii) If the executive owed \$47 000 at the beginning of July 2001 with interest payable at 7.8% per annum reducible and each fortnightly instalment was \$500, find in which year and month the loan will be repaid. [2 marks]

4

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} \quad (n \neq -1; x \neq 0 \text{ if } n < 0)$$

$$\int \frac{1}{x} dx = \log_e x \quad (x > 0)$$

$$\int \cos ax dx = \frac{1}{a} \sin ax \quad (a \neq 0)$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax \quad (a \neq 0)$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} \quad (a \neq 0)$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} \quad (a > 0, -a < x < a)$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log_e \left\{ x + \sqrt{x^2 - a^2} \right\} \quad (|x| > |a|)$$

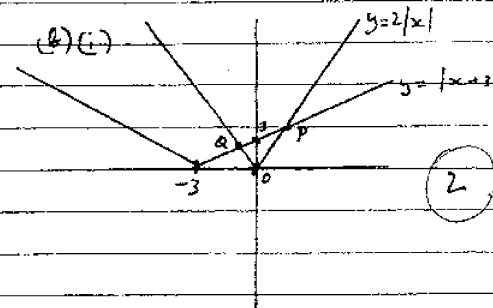
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log_e \left\{ x + \sqrt{x^2 + a^2} \right\}$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} \quad (a \neq 0)$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax \quad (a \neq 0)$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax \quad (a \neq 0)$$

$$\begin{aligned} (a) \int_0^{\pi/4} \cos 2x dx &= \frac{1}{2} \int_0^{\pi/4} (1 + \cos 4x) dx \\ &= \frac{1}{2} \left[x + \frac{\sin 4x}{4} \right]_0^{\pi/4} = \frac{\pi}{8} \end{aligned}$$



$$(i) \begin{cases} y = x+3 \\ y = 2x \end{cases} \Rightarrow 0 = x+3 \Rightarrow x = -3$$

$$P = (-3, 0)$$

$$\begin{cases} y = x+3 \\ y = -2x \end{cases} \Rightarrow 0 = 3x+3 \Rightarrow x = -1$$

$$Q = (-1, 2)$$

$$\therefore 2|x| \leq |x+3| \text{ if } -1 \leq x \leq 3$$

(c) Let a be fixed, d common difference

T_6, T_7, T_{10} geometric

$$\therefore (a+3d)^2 = (a+5d)(a+9d)$$

$$a^2 + 6ad + 9d^2 = a^2 + 14ad + 45d^2$$

$$\therefore 8ad + 36d^2 = 0$$

$$\therefore 4d(2a+9d) = 0$$

$$\therefore 2a+9d = 0 \text{ since } d \neq 0$$

$$(i) S_6 = \frac{6}{2} (2a+9d)$$

$$= 3(2a+9d) = 0$$

$$(ii) S_6 + S_{10} = \frac{6}{2} (2a+9d) + \frac{10}{2} (2a+11d)$$

$$= 6a + 15d + 12a + 66d$$

$$= 18a + 81d = 9(2a+9d) = 0$$

$$(iii) S_6 + S_6 - 2 \times S_{10} = 0$$

$$\therefore S_{12} - S_{10} = S_{10} - S_6$$

$$\therefore T_{12} + T_{11} = T_{10} + T_9 + T_8 + T_7$$

Extension 1 TRIAL 2001

A) (i) $\frac{dp}{dt} = k(p - 5000)$ — (1)

If $P = 5000 + Ae^{kt}$ — (2)

$\therefore \frac{dp}{dt} = Ake^{kt}$ — (3)

sub (2) and (3) into (1):

$\therefore \text{LHS} = Ake^{kt}$

$\text{RHS} = k(5000 + Ae^{kt} - 5000)$

$= Ake^{kt}$

$= \text{LHS}$

$\therefore P = 5000 + Ae^{kt}$ is a

solution of the differential

equation (1).

(ii) When $t=0$ $P=5002$

$\therefore 5002 = 5000 + Ae^0$

$\therefore A = 2$

$\therefore P = 5000 + 2e^{kt}$

when $t=6$ $P=25000$

$\therefore 25000 = 5000 + 2e^{6k}$

$\therefore 10000 = e^{6k}$

$\therefore k = \frac{1}{6} \ln 10000$

(iii) When $t=10$ $P=?$

$P = 5000 + 2e^{(\frac{1}{6} \ln 10000)10}$

$= 5000 + 2e^{\ln 10000^{10/6}}$

$= 5000 + 2(10000)^{10/6}$

$= 9288177.667...$

$\therefore$ mosquito population after

10 days is 9288178 (to

nearest mosquito).

(b) (i) $\ddot{x} = -n^2(x-b)$ — (1)

If $x = b - a \cos nt$ — (2)

$\therefore \dot{x} = an \sin nt$

$\ddot{x} = an^2 \cos nt$ — (3)

sub (2) and (3) into (1):

$\therefore \text{LHS} = an^2 \cos nt$

$\text{RHS} = -n^2(b - a \cos nt - b)$

$= -n^2(-a \cos nt)$

$= an^2 \cos nt$

$= \text{LHS}$

$\therefore x = b - a \cos nt$ satisfies the

equation for simple harmonic motion.

(ii)

Low Tide 9.25 am 7m

High Tide 3.40 pm $10\frac{1}{2}$ m

Period = $\frac{2\pi}{n}$

$\therefore 2 \times 6\frac{1}{2} = \frac{2\pi}{n}$

$\therefore n = \frac{2\pi}{25\frac{1}{2}} = \frac{4\pi}{25}$

$b = \frac{7 + 10\frac{1}{2}}{2} = 8\frac{5}{8}$

$a = \frac{10\frac{1}{2} - 7}{2} = 1\frac{5}{8}$

$\therefore x = 8\frac{5}{8} - 1\frac{5}{8} \cos \frac{4\pi}{25} t$

If $x = 9\frac{1}{2}$ $\therefore 9\frac{1}{2} = 8\frac{5}{8} - 1\frac{5}{8} \cos \frac{4\pi}{25} t$

$\therefore \frac{-2\frac{1}{2}}{1\frac{5}{8}} = \cos \frac{4\pi}{25} t$

$\therefore -\frac{4}{11} = \cos \frac{4\pi}{25} t$

$\therefore \cos^{-1} \left(-\frac{4}{11} \right) = \frac{4\pi}{25} t$

$\therefore t = \frac{25}{4\pi} \left[\pi - \cos^{-1} \left(-\frac{4}{11} \right) \right]$ for earliest time

$\therefore t = 3.865405773...$

$\therefore$ time after low tide is 3 hrs 52 mins (nearest min)

$\therefore$ earliest time for $9\frac{1}{2}$ m depth is 1.170 m

7) (a) TO PROVE: $3^n + 7^n$ is always even if $n \in \mathbb{N}^+$

PROOF: Step 1: When $n=1$ $3^1 + 7^1 = 3 + 7 = 10$, which is even

$\therefore$ it is true for $n=1$.

Step 2: Assume it is true for $n=k$ and prove it is true for $n=k+1$

ie $\frac{3^{k+1} + 7^{k+1}}{2} = m$ (where $m \in \mathbb{Z}$)

$\therefore 3^{k+1} + 7^{k+1} = 2m$ — (1)

If $n=k+1$ $3^n + 7^n = 3^{k+1} + 7^{k+1}$

$= 3 \cdot 3^k + 7 \cdot 7^k$

$= 3(2m - 7^k) + 7 \cdot 7^k$ (substituting (1))

$= 6m + 4 \cdot 7^k$

$= 2(3m + 2 \cdot 7^k)$, which is even.

$\therefore$ if it is true for $n=k$ so it is true for $n=k+1$.

Step 3: It is true for $n=1$ and so it is true for $n=1+1=2$. It is true for $n=2$ and so it is true for $n=2+1=3$ and so on for all positive integral values of n .

(b) (i) After 1 instalment the debt remaining $D_1 = P(1 + \frac{r}{100}) - F$

after 2 instalments the debt remaining $D_2 = D_1(1 + \frac{r}{100}) - F$

$= (P(1 + \frac{r}{100}) - F)(1 + \frac{r}{100}) - F$

$= P(1 + \frac{r}{100})^2 - F(1 + (1 + \frac{r}{100}))$

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$= P(1 + \frac{r}{100})^2 - F(1 + (1 + \frac{r}{100}))$

$$\therefore D_n = P\left(1 + \frac{r}{100}\right)^n - F \left[\frac{1 - \left(1 + \frac{r}{100}\right)^{-n}}{\frac{r}{100}} \right]$$

$$\therefore D_n = P\left(1 + \frac{r}{100}\right)^n - F \left[\frac{\left(1 + \frac{r}{100}\right)^n - 1}{\frac{r}{100}} \right], \quad \checkmark$$

$$(ii) \quad \text{Now if } D_n = 0 \quad \therefore P\left(1 + \frac{r}{100}\right)^n = F \left[\frac{\left(1 + \frac{r}{100}\right)^n - 1}{\frac{r}{100}} \right]$$

$$\therefore \frac{rP}{100} \left(1 + \frac{r}{100}\right)^n = F\left(1 + \frac{r}{100}\right)^n - F$$

$$\therefore F = \left(1 + \frac{r}{100}\right)^n \left[F - \frac{rP}{100} \right]$$

$$\therefore \left(1 + \frac{r}{100}\right)^n = \frac{F}{F - \frac{rP}{100}} \quad \checkmark$$

$$\therefore n \log_e \left(1 + \frac{r}{100}\right) = \log_e \left[\frac{F}{F - \frac{rP}{100}} \right]$$

$$\therefore n = \frac{\log_e \left[\frac{F}{F - \frac{rP}{100}} \right]}{\log_e \left(1 + \frac{r}{100}\right)}, \quad \checkmark$$

$$(iii) \quad P = 47000, \quad \frac{r}{100} = \frac{7.8}{100 \times 26} = 0.003, \quad F = 500$$

$$\therefore n = \frac{\log_e \left[\frac{500}{500 - 47000 \times 0.003} \right]}{\log_e [1 + 0.003]}$$

$$= 110.5941301 \dots$$

$\therefore$ The debt would be repaid in $\frac{110.5941301}{26} \dots$ years

$\Rightarrow$ 4 years and 6.59413 fortnights (to pay off loan).

$\therefore$ The loan will be repaid in October, 2005.